Mind the Gap: A Structured Survey of AI Mental Health Technologies and Their Equity Failures for Marginalized Populations

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***Abstract*—The rapid proliferation of AI-powered mental health technologies has produced a field defined by a foundational paradox: the populations most in need of accessible, affordable care are precisely those most likely to be harmed by current deployment practices. Queer, neurodivergent, disabled, racially and ethnically minoritized, and culturally diverse individuals face disproportionate barriers to traditional therapy and have turned to AI-driven conversational tools in substantial numbers. Yet systematic evidence reveals that these tools, designed predominantly on majority-population data and deployed without equity infrastructure, reproduce and often deepen the inequities they ostensibly address. This survey synthesizes evidence from 35 peer-reviewed sources published between 2020 and 2025, encompassing randomized controlled trials, systematic reviews and meta-analyses, observational studies, qualitative investigations, and policy analyses. Organized around three central questions (effectiveness, risk, and conditions for safe deployment), it identifies six converging patterns: purpose-built interventions outperform general-purpose tools; cultural adaptation and community co-design are safety requirements rather than optional enhancements; accessibility failures systematically exclude disabled users; crisis management infrastructure lags dangerously behind deployment; Large Language Model (LLM) demographic bias is empirically documented and pervasive; and safety standards remain underdeveloped relative to scale. The survey concludes by proposing an equity-centered design framework and identifying five critical gaps that the research community must urgently address.**

***Index Terms*—AI mental health, marginalized populations, chatbot equity, digital mental health, LGBTQ+ mental health, neurodivergent users, racial bias, cultural adaptation, accessibility, algorithmic bias, crisis management, co-design, literature survey**

I. Introduction

Mental health disorders represent a leading cause of disability worldwide, yet the populations who bear the greatest burden of psychological distress are systematically excluded from the care systems meant to serve them [1]. Sexual and gender minorities experience depressive symptoms at 1.5 times the general population rate, with 59% of transgender individuals experiencing clinically significant depression and 40% of Lesbian, Gay, Bisexual, Transgender, and Queer (LGBTQ+) youth seriously considering suicide [2]. Blind individuals face elevated rates of depression and anxiety while encountering systematic exclusion from mental health services [3]. Neurodivergent individuals, including autistic people and those with Attention Deficit Hyperactivity Disorder (ADHD), experience mental health challenges at substantially elevated rates while finding traditional therapeutic modalities poorly adapted to their cognitive and communicative profiles [4]. Racially and ethnically minoritized communities face documented bias in clinical AI systems that threatens to replicate and amplify existing healthcare disparities [5], [6]. Culturally and linguistically diverse communities struggle with systems that fail to recognize cultural expressions of distress as clinically meaningful [7].

AI-powered mental health technologies have emerged with considerable promise into this landscape. They offer on-demand interaction, anonymity, low cost, and no appointment requirements, properties that are particularly valuable for populations for whom stigma and cost constitute primary barriers [8], [9]. Meta-analytic evidence confirms that AI-based conversational agents produce significant reductions in depression (Hedges' $g = 0.64$) and psychological distress (Hedges' $g = 0.70$) [8], [10]. The field is expanding rapidly: a bibliometric analysis found a 46.19% annual growth rate in chatbot mental health publications between 2015 and 2024 [11].

Yet rapid expansion has not been accompanied by adequate equity infrastructure. General-purpose chatbots repurposed for mental health have been shown to harm users with depression [12]. Cultural bias causes AI systems trained on Western data to fail consistently for non-Western populations [13]. Accessibility barriers systematically exclude disabled users from platforms that ostensibly expand access to care [3].

Crisis recognition failures place vulnerable users at risk during acute emergencies [14]. Demographic bias in LLMs is now empirically documented at scale across race, gender, and ethnicity [15], [16]. Socioeconomic barriers compound each of these risks, ensuring that the populations who turn to AI tools out of financial necessity are also those least likely to receive adequate care [17], [5]. The American Psychological Association has issued explicit guidance warning against generic chatbot use for mental health support [18], and the World Health Organization (WHO) has called for robust ethics and governance frameworks for large multimodal models in health contexts [19].

This survey synthesizes the evidence at this inflection point, where deployment has outpaced safety and equity lags behind scale. It addresses three central research questions:

- **RQ1:** How effective are AI-powered mental health interventions for marginalized populations?
- **RQ2:** What unique risks do marginalized communities face when using AI mental health technologies?
- **RQ3:** Under what conditions can AI mental health technologies be safely and effectively deployed for marginalized populations?

The survey contributes by synthesizing evidence across population groups rather than within single identity categories, analyzing equity implications of deployment at scale, identifying design principles grounded in empirical evidence, and surfacing critical research gaps that the field must address as a matter of urgency.

II. Survey Methodology

A. Search Strategy

This survey employed a systematic search procedure across seven databases: ACM Digital Library, PubMed, PsycINFO, IEEE Xplore, MEDLINE, Scopus, and Google Scholar. The search strategy combined three concept groups using Boolean operators: AI and digital mental health terms (*chatbot, conversational agent, large language model, digital mental health*); marginalized population identifiers (*LGBTQ+, disabled, neurodivergent, autistic, ADHD, culturally diverse, non-Western, racial minority, low-income*); and outcome measures (*depression, anxiety, safety, equity, bias, accessibility*). Additional literature was identified through citation searching of included studies and relevant systematic reviews.

B. Inclusion and Exclusion Criteria

Studies were included if they: examined AI-powered or digital mental health interventions; included marginalized or vulnerable populations as primary samples or subgroup analyses; reported empirical data from Randomized Controlled Trial (RCT)s, observational studies, mixed-methods research, or systematic reviews; were published in peer-reviewed venues between 2020 and 2025; and addressed effectiveness, safety, accessibility, or equity considerations. Studies were excluded if they focused solely on general populations without marginalized group analysis, were purely theoretical without empirical data, examined non-AI digital interventions, were conference abstracts or editorials, or were non-peer-reviewed grey literature.

C. Corpus Composition

Thirty-five studies met inclusion criteria. The corpus encompasses five RCTs [12], [20], [2], [21], [22]; seven systematic reviews and meta-analyses [10], [1], [23], [8], [2], [15], [6]; five observational studies [14], [9], [3], [24], [4]; five qualitative and mixed-methods studies [13], [25], [26], [7], [27]; one implementation science study [28]; two policy and institutional reports [18], [19]; five additional systematic reviews and critical analyses [29], [30], [16], [5], [17]; one bibliometric analysis [11]; and four complementary studies [31], [32], [33], [34].

D. Synthesis Approach

Following Webster and Watson's [35] guidance on concept-centric literature review methodology, analysis employed thematic synthesis organized around the three research questions. Studies were categorized by intervention type and population focus. A concept matrix (Table I) organizes the literature by theme, exposing cross-cutting patterns that author-by-author review obscures.

Two seminal works anchor the survey. Khan and Seo [3] provide the first comprehensive empirical documentation of Digital Mental Health (DMH) accessibility failures for blind users, establishing accessibility as a safety issue rather than a design preference. Karkosz et al. [20] provide one of the first rigorous RCTs demonstrating both promise and limitations of culturally-adapted AI mental health interventions in a non-English language, establishing an evidence-based methodology for cross-cultural adaptation.

III. Effectiveness of AI Mental Health Interventions

A. Purpose-Built Interventions: Consistent Evidence of Efficacy

The effectiveness landscape reveals a clear structural distinction: purpose-built mental health tools produce meaningful outcomes, while general-purpose AI repurposed for mental health produces negligible gains and may cause harm.

A RCT of Fido, a Polish-language Cognitive Behavioural Therapy (CBT) chatbot designed specifically for non-English speakers, demonstrated depression score reductions of 8 points

and anxiety reductions of 3 points, with effects stable at one-month follow-up [20]. Participants achieved outcomes faster than controls (79.4 vs. 117.6 minutes, $p = 0.008$, $d = 0.64$), and the chatbot group showed substantially lower attrition (10% vs. 37%, $p < 0.001$) [20]. A real-world study of Wysa with over 2,000 users demonstrated significant improvements for chronic pain patients, with depression declining from 13 to 10 and anxiety from 12 to 9; users with chronic pain showed higher engagement than general users, with 67.7% of user-provided feedback indicating the experience was helpful [21].

Meta-analytic evidence confirms these patterns at scale. Li et al. [8] synthesized 35 studies (15 RCTs included in meta-analysis) and found significant reductions in depression (Hedges' $g = 0.64$, 95% CI [0.17, 1.12]) and psychological distress (Hedges' $g = 0.70$, 95% CI [0.18, 1.22]). Effects were stronger for multimodal interventions ($g = 0.89$ vs. 0.45 text-only), generative AI ($g = 0.78$ vs. 0.52 rule-based), and mobile-integrated formats ($g = 0.73$ vs. 0.48 web-only) [8]. An earlier meta-analysis corroborated these findings but rated evidence quality as "weak" due to small sample sizes and high risk of bias in 58% of included studies, with only two studies assessing safety [10].

B. LGBTQ+ Populations

For LGBTQ+ populations, a systematic review of 15 digital interventions identified that structured formal interventions and professionally-facilitated mobile applications most consistently reduced symptoms [2]. Programs like AFFIRM Online demonstrated significant depression reduction. Applications that explicitly acknowledged limitations and communicated clearly when users should seek additional help achieved better outcomes than those claiming comprehensive support [9]. Critical gaps constrain generalizability: few interventions included LGBTQ+ youth of colour, most research occurred in high-income Western nations, and non-binary and gender non-conforming youth remained severely underrepresented [2].

C. Neurodivergent Populations

Neurodivergent individuals represent a critically underexamined population in the DMH literature despite compelling evidence of unmet need. Autistic people and those with ADHD experience mental health conditions, including depression and anxiety, at rates two to four times those of the general population, yet face systematic barriers in traditional therapeutic settings, including sensory sensitivities, communication differences, and the cognitive overhead of masking neurotypical behavior [4]. AI chatbots offer properties specifically appealing to this community: consistent, patient, non-judgmental interaction without the unpredictability of human social dynamics [4].

A CHI 2025 study examining autistic individuals' perspectives on LLM-based support for negative self-talk found that, while many autistic users already rely on LLMs for mental health guidance, mental health practitioners identified significant concerns about neurotypical bias in AI responses [27]. Practitioners critiqued LLM outputs as "overly wordy, vague, and overwhelming," reflecting assumptions about how neurotypical individuals process information [27]. A 2025 commentary on AI companionship for autistic people documented that companion AI systems present a unique benefit (a low-pressure, judgment-free social environment) and a unique danger (insufficient crisis detection and the risk of reinforcing social isolation rather than complementing human connection) [4]. A scoping review of generative AI for autistic populations across screening, assessment, and intervention found consistent performance improvements over baseline methods but also documented small sample sizes, data biases, model hallucinations, and limited validation across diverse autistic presentations [34].

D. Racial and Ethnic Minority Populations

The racial and ethnic mental health equity literature reveals a compounding dynamic: existing clinical disparities, when encoded into AI training data, are reproduced and potentially amplified at scale. A systematic review of 30 studies found a significant association between AI utilization and the exacerbation of racial disparities, with Black and Hispanic populations showing the most consistent patterns of harm [6]. Prior research documents that AI models trained predominantly on majority-population data are less likely to recommend advanced care for patients from underrepresented racial groups, even when controlling for clinical indicators [5].

In mental health contexts specifically, Timmons et al. [5] conducted a comprehensive review of health equity implications in AI mental health applications, documenting that racially and ethnically minoritized groups are less likely to seek and obtain treatment, more likely to receive poor quality services when treated, and least likely to be represented in the training datasets that shape AI systems. Their "fair-aware AI" framework proposes that bias assessment must be embedded across the full model development lifecycle, from data collection through deployment monitoring [5]. Straw and Callison-Burch [33] demonstrated that language models exhibit systematic racial and gender bias in mental health contexts, with outputs reflecting and reinforcing stereotypes that could directly harm clinical decision-making [33].

E. Socioeconomically Marginalized Populations

Socioeconomic status represents a structural risk factor that cuts across all population groups surveyed here. Robinson et al. [17] document that individuals from resource-limited backgrounds are among the fastest-rising adopters of smartphones, with 76% owning a device, yet simultaneously face the greatest barriers to high-quality DMH tools due to cost, digital literacy gaps,

limited connectivity, and the absence of culturally responsive content [17]. The paradox is acute: the populations most likely to turn to AI mental health tools out of financial necessity are those least served by tools designed for and tested on higher-income, higher-literacy, majority-population users.

Huang et al. [31] document that ethnically diverse young people show significantly lower engagement with DMH interventions than majority-population users in randomized trials, suggesting that even when interventions are made available, their design systematically fails to engage the populations most in need [31].

F. Generative AI: Complexity and Caution

Recent evidence on generative AI presents complexity that demands careful interpretation. A UK study found conversational AI facilitated mental health assessments and improved recovery rates when used for screening and triage with human oversight [22]. In contrast, a three-arm RCT comparing a specialized mental health chatbot with a general chatbot found the general chatbot condition produced concerning safety signals: the specialized chatbot group showed significant depression reduction (PHQ-9 change: -2.44, $p < 0.001$), while the general chatbot group showed minimal improvement (PHQ-9 change: -0.63, $p = 0.28$), with five participants requiring human intervention due to elevated risk [12]. A comprehensive systematic review across mental health AI contexts found that ethical considerations and methodological limitations warrant urgent attention before widespread deployment [23].

G. Cultural Adaptation as Efficacy Determinant

Cultural adaptation determines whether interventions succeed or fail, and its absence constitutes both an efficacy failure and an equity harm. A narrative review established that surface-level adaptations prove insufficient without deep-structure adaptations addressing how communities perceive illness causes, treatment expectations, and help-seeking norms [26]. A qualitative study with Culturally and Linguistically Diverse (CALD) young people revealed chatbots misunderstanding cultural expressions of distress and providing advice incompatible with collectivist family structures [7]. A study examining ChatGPT found consistent failures when presented with mental health scenarios in non-Western cultural frameworks: accuracy dropped to 34% for collectivist scenarios versus 78% for individualist Western scenarios, with the model misunderstanding family obligations and community responsibilities in 66% of non-Western test cases [13].

IV. Risks Facing Marginalized Communities

A. Accessibility Exclusion

Accessibility exclusion constitutes the most structurally entrenched risk. Platforms that ostensibly expand care actively block disabled users from accessing it. A study with 93 blind users documented systematic accessibility failures across DMH platforms despite legal requirements and available technical solutions [3]. Most DMH tracking services lacked screen reader compatibility, used visual-only displays without text alternatives, and created interaction patterns impossible to navigate without sight. Participants described forced choices between inaccessible mainstream applications with full features or specialized disability-focused platforms with limited functionality [3]. This affects a population that already experiences depression and anxiety at two to three times the rates of sighted individuals.

The pattern mirrors decades of web accessibility failure: legal requirements exist, implementation is ignored, and disabled populations are structurally locked out of technologies claimed to democratize access [3].

B. Demographic Bias in Large Language Models

LLM demographic bias is empirically documented. Omar et al. [15] conducted a systematic review of 24 studies evaluating demographic biases in medical LLMs and found that 22 (91.7%) identified biases, with gender bias reported in 15 of 16 studies (93.7%) and racial or ethnic biases in 10 of 11 studies (90.9%). Pfohl et al. [16] developed a multifactorial framework for assessing equity-related LLM harms and conducted a large-scale empirical study with the Med-PaLM 2 LLM, involving 806 raters from diverse demographic backgrounds and generating over 17,000 human ratings; their findings revealed systematic bias patterns that narrower evaluation approaches missed. Lawrence et al. [29] identified additional dimensions of LLM risk in mental health contexts, including the absence of standardized evaluation protocols and compounding effects when LLMs are deployed at scale.

These findings carry particular weight for mental health applications. In clinical presentations, race, gender, sexuality, disability status, and socioeconomic status are directly implicated in symptom expression, help-seeking behavior, treatment response, and likelihood of accurate diagnosis. Bias on these dimensions in mental health AI carries direct clinical consequences [5], [6].

Straw and Callison-Burch [33] specifically examined language model bias in mental health contexts, demonstrating that general-purpose LLMs reproduce racial and gender stereotypes in mental health-relevant outputs, providing a mechanistic account of how bias encoded in training data propagates into care-relevant advice. The broader review by Haider et al. [6] corroborates this at scale across 30 studies, finding consistent racial disparities in AI-driven healthcare recommendations.

C. Neurotypical Bias Against Neurodivergent Users

A distinct category of bias affects neurodivergent users: neurotypical bias, in which LLMs trained predominantly on neurotypical-authored text encode assumptions about emotional processing, social communication, and help-seeking that may conflict with neurodivergent experience [27]. Responses considered empathic and clear by neurotypical raters were judged by clinicians working with autistic populations as "overly wordy, vague, and overwhelming," reflecting a mismatch between assumed and actual cognitive needs [27]. For a population already prone to misunderstanding in clinical encounters, AI systems that replicate these misunderstandings at scale represent a structural barrier to care [4].

Additionally, the risk of emotional over-reliance is particularly pronounced for autistic users. Papadopoulos [4] documents cases in which autistic individuals developed primary social bonds with AI companions, to the detriment of human connection, and in one documented case a companion AI failed to intervene during acute suicidal ideation, reflecting the same crisis detection failure documented across populations by De Freitas et al. [14].

D. Online Safety and Harassment

Online safety risks disproportionately burden LGBTQ+ youth using digital platforms for mental health support. An analysis of 173 youth Instagram users (86 LGBTQ+, 87 heterosexual) revealed that LGBTQ+ youth experienced significantly more high-risk online interactions: 68% experienced sexual messages from strangers compared to 23% of heterosexual youth ($p < 0.001$), and 34% experienced online interactions directly linked to self-harm behaviors compared to 11% of heterosexual youth ($p < 0.001$) [24]. Each additional high-risk interaction category correlated with 1.4-point increases in PHQ-9 depression scores and 1.8-point increases in Generalized Anxiety Disorder 7-item Scale (GAD-7) anxiety scores [24]. Existing AI content moderation demonstrated dual failures: only 23% of reported harassment resulted in content removal, while 41% of identity-affirming posts were flagged as inappropriate [24].

E. Crisis Mismanagement and Harmful Advice

Crisis recognition and response represent the most acutely dangerous risk domain. Mental health crises appeared in 17% of conversations with companion AIs ($n = 289$), including suicidal ideation (8%), severe depression (12%), and self-harm discussion (9%) [14]. Performance testing found chatbots failed to recognize crisis situations in 62% of test scenarios involving suicidal ideation and provided appropriate escalation responses in only 31% of detected crises [14]. In 23% of crisis conversations, rigid safety guardrails rejected users during moments of vulnerability with generic responses, potentially increasing distress rather than mitigating it [14].

The real-world stakes of crisis mismanagement are illustrated by the case of National Eating Disorders Association (NEDA)'s Tessa chatbot. Deployed as a peer support tool, Tessa was shut down in May 2023 after users reported it provided harmful weight-loss advice, recommending calorie restriction and weight tracking in direct contradiction to evidence-based eating disorder treatment [36]. The case exemplifies De Freitas et al.'s empirical findings: AI trained on general data lacks the clinical knowledge to navigate complex mental health situations safely, and the assumption that accessibility automatically benefits vulnerable users is demonstrably false [14].

Palmer et al. [32] add a regulatory dimension, documenting that none of the three dominant regulatory models for digital mental health tools (laissez-faire, highly regulated, and the current Food and Drug Administration (FDA) approach) satisfactorily balances the promise of access with ethical risk management, leaving a structural gap through which harmful deployments continue to reach vulnerable populations [32].

F. Ethical Violations in Therapeutic AI Deployment

Meadi et al. [30] systematically examined the ethical challenges of deploying conversational AI "as therapist." Their analysis identified violations across autonomy, beneficence, non-maleficence, and justice, finding that platforms routinely overclaim therapeutic capability, inadequately disclose limitations, and fail to provide meaningful alternatives when users require escalated care. Marginalized populations bear elevated exposure to these ethical failures because they have fewer alternatives when AI interventions fail [30].

G. Implementation Gaps and Sustainability Risks

Implementation gaps create systemic vulnerabilities. An expert panel review identified that current AI mental health tools lack standardized safety testing protocols, real-time monitoring systems, clear crisis escalation pathways, long-term outcome tracking, and mechanisms to identify when users discontinue due to harm [28]. Dehbozorgi et al. [23] corroborate these findings, establishing that ethical frameworks and safety protocols remain underdeveloped relative to deployment scale. A scoping review additionally identified sustainability crises, with chatbot companies lacking viable business models: when platforms shut down, vulnerable users face abrupt service termination with no continuity of care and no human alternative [1].

V. Conditions for Safe, Effective Deployment

Six design principles emerge from the evidence as necessary conditions for equity-centered deployment.

A. Community Co-Design as Safety Imperative

Co-design with target communities functions as both an efficacy requirement and a safety requirement. Only half of LGBTQ+

interventions included meaningful community input, yet those employing co-design demonstrated higher effectiveness, better engagement, and greater trust [25]. Community involvement helped identify invisible harms, incorporate affirming language, and build the credibility that drives sustained engagement [25]. Outsider-developed interventions provided advice immediately recognizable as culturally inappropriate by anyone with lived experience [7]. For neurodivergent populations, Papadopoulos [4] specifically emphasizes that co-design with neurodivergent developers and users produces qualitatively different tools from those designed by neurotypical teams alone.

B. Cultural Humility Through Continuous Adaptation

Static translation fails; culturally humble approaches require continuous community engagement. Naderbagi et al. [26] established a framework distinguishing surface-level adaptation from deep-structure adaptation that addresses how populations conceptualize mental health, perceive illness causes, define treatment success, and approach help-seeking. The framework requires iterative testing with diverse users, flexibility accommodating within-group diversity, and recognition that culture extends beyond ethnicity to encompass family structure, community norms, and historical experiences of healthcare systems.

C. Hybrid Human-AI Models

Hybrid architectures balance scalability with safety. Conversational AI effectively facilitates assessments and triage when human oversight is embedded in the system architecture [22]. Professionally-facilitated digital programs achieve both higher engagement and better safety outcomes than fully automated systems [2]. Human escalation pathways must be built in from inception [28]. For high-risk populations, including those with active suicidal ideation or severe trauma histories, full automation represents an inadequate standard of care regardless of chatbot sophistication [14], [32].

D. Transparency About Limitations

Transparency improves outcomes and is a precondition for informed consent. Explicitly acknowledging limitations produces better results than claiming comprehensive support [9]. Smith et al. [28] recommend mandatory disclosure covering what conditions chatbots address, what evidence supports effectiveness, what safety protocols exist, what data privacy protections apply, and what chatbots cannot do. The American Psychological Association has reinforced this position, recommending against generic chatbot use for mental health support [18]. Palmer et al. [32] propose a voluntary certification program that would standardize disclosure requirements across digital mental health platforms.

E. Accessibility as Foundational Architecture

Accessibility must be foundational from the earliest design stages. Khan and Seo [3] establish that genuine accessibility requires involving disabled users throughout development, treating Web Content Accessibility Guidelines (WCAG) compliance as a minimum rather than a ceiling, conducting testing with actual assistive technology users, and designing interaction patterns that work across diverse ability levels by default. For neurodivergent users, accessible design additionally requires attention to cognitive load, sensory sensitivity, communication style variability, and the absence of neurotypical social cues in interaction design [4], [27].

F. Fair-Aware AI Development

Timmons et al. [5] propose a "fair-aware AI" framework requiring bias assessment at every stage of the model development lifecycle, from training data curation and feature selection through model evaluation, deployment monitoring, and post-market surveillance. Their framework specifically identifies that mental health AI must be evaluated against equity outcomes (whether it reduces or widens disparities) alongside efficacy outcomes (whether it reduces symptoms) [5]. This framework extends to racial and ethnic bias [6], [15], gender and sexuality bias [33], and neurotypical bias [27], providing a unified methodological approach to equity across population groups.

VI. Thematic Analysis

A. Concept Matrix

Table I organizes the surveyed literature by theme following Webster and Watson's [35] concept-centric methodology, revealing patterns that author-by-author review obscures.

B. The Trapped User Phenomenon

The evidence across themes reveals a structural paradox this survey terms the *trapped user phenomenon*: marginalized individuals turn to digital mental health after traditional services fail due to cost, stigma, or cultural incompetence, yet have no recourse when AI fails them in turn. Khan and Seo's blind participants described having "no choice" while sighted users have their "pick of the litter" [3]. De Freitas et al. [14] documented users disclosing suicidal ideation to chatbots that could not recognize or respond to it. Papadopoulos [4] documents autistic individuals developing primary social bonds with AI companions because no accessible human alternative existed. Timmons et al. [5] frame this as a structural inequity: the communities most likely to turn to AI mental health tools out of necessity are also those whose data is least represented in training corpora, whose clinical needs are least reflected in design decisions, and whose harms are least captured in post-market surveillance.

TABLE I
CONCEPT MATRIX: AI MENTAL HEALTH TECHNOLOGIES AND MARGINALIZED POPULATIONS

Theme	Evidence of Effectiveness	Evidence of Risks/Harms	Cultural/Accessibility Considerations	Design Recommendations
Purpose-Built CBT Chatbots	[20]: depression ↓8 pts; [21]: depression $13 \rightarrow 10$; [8]: Hedges' $g = 0.64$	[12]: general chatbots may harm; [14]: crisis recognition failures	[20]: culturally adapted for Polish speakers; [21]: chronic pain populations show higher engagement	[22]: hybrid human-AI models; [28]: safety protocols required
LGBTQ+ Populations	[2]: structured formal interventions reduce symptoms; [9]: m-health apps show promise	[24]: elevated online harassment and self-harm links; [25]: only 50% include community input	[2]: underrepresentation of non-binary/trans youth of colour; [25]: identity-affirming content critical	[2], [25]: community co-design essential; transparency about limitations
Neurodivergent Populations	[4]: judgment-free interaction benefits documented; [34]: AI screening improvements	[27]: neurotypical bias in LLM outputs; [4]: crisis detection failures; over-reliance risk	[27]: responses overwhelm cognitive load; [4]: social isolation risk	[4]: co-design with neurodivergent developers; [27]: specialist practitioner oversight
Racial/Ethnic Minorities	— (significant gap in purpose-built tools)	[6]: AI exacerbates racial disparities; [5]: underrepresentation in training data; [33]: racial stereotypes in LLM outputs	[5]: minoritized groups less likely to seek treatment; [6]: Black and Hispanic populations most affected	[5]: fair-aware AI framework; [15]: diverse rater pools in evaluation
Cultural Adaptation	[26]: culturally adapted interventions more effective	[13]: ChatGPT fails non-Western contexts; [7]: misunderstood cultural distress	[7]: CALD youth need culturally humble approaches; [26]: deep vs. surface adaptation distinction	[26]: continuous community engagement; [7]: avoid one-time translation
Accessibility for Disabled Users	[3]: documents gap (most DMH platforms inaccessible)	[3]: screen reader failures; visual-only displays exclude blind users	[3]: forced choice between inaccessible mainstream or limited disability-specific tools	[3]: accessibility as foundational design requirement
Socioeconomic Barriers	[17]: smartphone adoption high among low-income users	[17]: digital divide limits equitable access; [31]: diverse youth show lower engagement	[17]: low-income users depend on phone-only access; digital literacy gaps	[17]: DEIB in full product lifecycle; [5]: equitable evidence generation
LLM Demographic Bias	— (evaluation frameworks beginning to emerge)	[15]: bias in 91.7% of medical LLM studies; [16]: equity harm framework; [33]: racial/gender bias in mental health contexts	[15]: gender bias 93.7%, racial bias 90.9% of studies; [6]: Black/Hispanic populations most affected	[16]: diverse rater pools; [5]: bias assessment throughout lifecycle
Safety & Crisis Management	[10]: weak evidence of safety in early literature	[14]: crisis recognition failures; [12]: harmful advice documented; [36]: Tessa case	[28]: lack of crisis escalation protocols; [32]: regulatory gap	[28]: human oversight mandatory; [32]: voluntary certification program
Implementation & Sustainability	[1]: scoping review identifies gaps	[23]: implementation challenges; lack of long-term monitoring	[28]: no standardized testing; limited post-discontinuation tracking	[8]: multimodal, mobile-integrated apps more effective; [28]: safety standards pre-deployment

The trapped user phenomenon explains why harm from AI mental health tools is not evenly distributed: it concentrates among those who had nowhere else to go.

C. Stratified Digital Access

The evidence challenges techno-optimist narratives that position digital mental health as inherently democratizing. Availability does not equal accessibility [3]. Global deployment means little when cultural inaccessibility renders AI useless for the populations it reaches [13]. A more accurate model is "stratified digital access": AI systems trained on majority populations and deployed without equity design create bifurcated landscapes where privileged groups receive adequate support while marginalized populations receive inadequate or harmful support with no human alternatives [17], [5].

The WHO's 2025 guidance on AI ethics in health [19] reinforces this critique at the policy level, emphasizing that governance frameworks must address equity from the outset of system design. Robinson et al. [17] call on manufacturers, academic researchers, payers, providers, and regulatory agencies to collectively specify and monitor equity requirements before adopting or deploying DMH products.

VII. Research Gaps and Future Directions

Five critical gaps require urgent attention.

Long-term outcome tracking is nearly absent. Smith et al. [28] establish that the field lacks evidence about sustained benefits, delayed harms, or consequences when users discontinue chatbot use. Without longitudinal data, clinical claims cannot be evaluated against real-world outcomes.

Intersectionality receives minimal attention. Studies examine single identity categories rather than multiply-marginalized individuals who face compounded barriers. LGBTQ+ individuals who are also disabled, non-Western, and low-income, or neurodivergent individuals who are also racially minoritized, represent the most underserved populations, yet they appear in almost no research [2], [5].

Crisis performance is dangerously understudied. Limited systematic evidence exists about AI performance during acute suicidal emergencies, precisely the context where failure proves fatal [14]. Existing evidence from companion AI and general chatbot studies is concerning; purpose-built crisis response has received almost no controlled investigation [28].

Neurodivergent-specific design remains nascent. Despite compelling evidence of unmet need and specific design requirements, the field lacks RCTs of purpose-built AI mental health tools for autistic or ADHD populations and lacks standardized evaluation protocols for neurotypical bias [27], [4], [34].

Comparative effectiveness research comparing AI with appropriately matched human psychotherapy for marginalized populations does not exist at the scale necessary to make evidence-based deployment recommendations [10].

VIII. Policy Implications

The evidence supports five urgent policy interventions.

Mandatory pre-deployment safety standards should require standardized testing for crisis recognition, cultural bias, neurotypical bias, accessibility, data privacy, and adverse event monitoring before any AI tool is deployed for vulnerable populations. The current absence of such requirements, documented by Smith et al. [28] and reinforced by the NEDA case [36], represents a structural failure of regulatory oversight.

Community co-design certification should mandate documented community engagement processes, cultural adaptation evidence, accessibility testing with disabled users, neurodivergent user testing, and ongoing advisory boards for any AI tool claiming to serve marginalized populations.

Transparency and informed consent should provide users with clear information about what chatbots address, what evidence supports effectiveness, what limitations apply, when human support is necessary, how data is used, and what crisis escalation exists. Palmer et al. [32] propose a voluntary certification program as one practical mechanism for standardizing these disclosures.

Hybrid model prioritization for high-risk populations should require human oversight for any AI deployment serving populations with elevated suicide risk or acute mental health vulnerability, with particular attention to neurodivergent and LGBTQ+ youth.

Accessibility as infrastructure should require platforms to meet WCAG standards and demonstrate usability across assistive technologies before deployment, treating accessibility as a prerequisite for service provision [3].

IX. Conclusion

AI mental health technologies demonstrate both remarkable therapeutic potential and serious safety risks for marginalized communities. Purpose-built interventions show that culturally-adapted, evidence-based AI can deliver meaningful benefits to underserved populations, with meta-analytic evidence confirming significant symptom reductions [10], [8]. However, systematic accessibility failures [3], pervasive demographic bias in LLMs [15], [16], neurotypical bias against neurodivergent users [27], cultural misunderstanding [13], racial disparities in AI healthcare outcomes [6], [5], online safety crises [24], harmful advice and crisis mismanagement [14], [12], [36], and implementation gaps [28] together demonstrate that current deployment patterns deepen rather than reduce mental health inequities.

The stakes are concrete. When AI mental health tools fail, people suffer. The populations surveyed here, spanning LGBTQ+ youth, neurodivergent adults, blind individuals, racially and ethnically minoritized communities, culturally diverse young people, and those with limited economic resources, face elevated mental health risk, systematic barriers to traditional care, and histories of healthcare discrimination. They deserve better than being unpaid beta testers for technology deployed without adequate safety protocols. This survey points toward a path forward grounded in three principles: equity must be designed in from conception through deployment [25], [3], [26]; safety infrastructure, including crisis protocols, human

oversight, accessibility standards, and outcome monitoring, must precede deployment [14], [28]; and transparency, with clear information about limitations, easy access to human alternatives, and meaningful data control, must be preserved [9], [22].

The promise of AI mental health technologies for marginalized populations is real. Realizing that promise requires confronting current reality: existing approaches too often fail precisely those populations most in need. The future of equitable digital mental health depends on whether the field can move from rapid deployment to responsible development, from techno-optimism to evidence-based implementation, and from designing for imagined users to co-creating with real communities.

List of Acronyms

ADHD Attention Deficit Hyperactivity Disorder 1, 3, 8, 9

AI Artificial Intelligence 9

CALD Culturally and Linguistically Diverse 4, 9

CBT Cognitive Behavioural Therapy: A structured, evidence-based psychotherapeutic approach targeting the relationship between thoughts, feelings, and behaviours; the most common modality delivered by purpose-built mental health chatbots 2, 9

DMH Digital Mental Health: The field encompassing mobile applications, online platforms, chatbots, and AI-powered tools designed to deliver mental health support outside traditional clinical settings 2–4, 8, 9

FDA Food and Drug Administration 5, 9

GAD-7 Generalized Anxiety Disorder 7-item Scale 5, 9

LGBTQ+ Lesbian, Gay, Bisexual, Transgender, and Queer 1, 3, 5, 8, 9

LLM Large Language Model 1–5, 8, 9

NEDA National Eating Disorders Association 5, 8, 9

RCT Randomized Controlled Trial 2–4, 8, 9

WCAG Web Content Accessibility Guidelines: The internationally recognized technical standards for digital accessibility, published by the World Wide Web Consortium, specifying minimum requirements for screen reader compatibility, colour contrast, and interaction design 6, 8, 9

WHO World Health Organization 2, 8, 9

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